SYSTEMS ANALYSIS AND DESIGN · ITC327W

Click Computer and Accessories

Client Support Platform

Phase 1 Presentation · Group T · September 2026

GROUP T — TEAM MEMBERS

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ROADMAP

Agenda

1. Problem Statement
2. Stakeholders
3. Current Process
4. Requirements — Functional & Non-Functional
5. Proposed Solution & Architecture
6. Feasibility Study
7. Risk Analysis
8. Project Schedule (MS Project)
9. Key Decisions & Scope Changes
10. Team Contributions
11. Conclusion & Q&A

THE CHALLENGE

Problem Statement

Current Situation

- Clients contact the business directly by phone or in person
- No centralised system for tracking support requests
- Manual assignment of technicians
- Limited visibility for clients

Specific Problem

- Lack of a simple, centralised platform for:
- Submitting support requests
- Tracking request progress
- Managing assignments and communication

Consequences

- Delays in receiving support
- Requests being missed or forgotten
- Limited visibility of progress
- Increased workload for support staff

A centralised platform removes the guesswork — every request tracked, every technician accountable.

WHO'S INVOLVED

Stakeholders

Prince Morafo	*Click Computer & Accessories — External Stakeholder / Business Owner*	Provides requirements and sign-off
Clients	*End Users*	Submit and track support requests
Support Operators	*Internal Staff*	Monitor and manage incoming requests
Technicians	*Internal Staff*	Handle assigned technical problems
Administrators	*Internal Staff*	Manage users and service operations

KEY STAKEHOLDER NEEDS

- Simple client application
- Request submission and tracking
- 24/7 active management
- Technician assignment
- Administrator controls

Five stakeholder groups, one shared goal: a faster, more visible support process.

AS-IS

Current Process

Client → Contact Staff → Manual Notes → Assign Technician → Repair → Feedback

No tracking system exists at any stage of this process

PROCESS PROBLEMS

- Delays in receiving support
- Difficulty communicating technical problems clearly
- Requests being missed or forgotten
- Difficulty assigning to appropriate technicians
- Limited visibility of request progress
- Increased workload for support staff
- Poor communication

REQUIREMENTS

Functional Requirements

FR1	Client registration and login	Must Have
FR2	Device menu (Cellphone, Laptop, Printer, TV, Other)	Must Have
FR3	Problem / service selection	Must Have
FR4	Request submission with description	Must Have
FR5	Request status tracking	Must Have
FR6	Client / staff communication	Must Have
FR7	Technician assignment	Must Have
FR8	Administrator controls	Must Have
FR9	Notifications	Should Have
FR10	Asset / device information	Should Have
FR11	Supporting evidence (screenshots, photos, documents)	Should Have

11
Total FRs

8
Must Have

3
Should Have

REQUIREMENTS

Non-Functional Requirements

S	**NFR1**	Protect user information from unauthorised access
✓	**NFR2**	Clear confirmation, status, and error messages
≡	**NFR3**	Accurate and consistent request information
→	**NFR4**	Simple navigation with minimal steps for submission
	NFR5	API responses within 2 seconds under normal load

5

Non-Functional Requirements

All Must Have

THE APPROACH

Proposed Solution & Architecture

TECHNOLOGY STACK

Frontend	Flutter (Dart)
Backend API	ASP.NET Core 6.0 (C#)
Database	Supabase (PostgreSQL)
Authentication	Supabase Auth
Storage	Supabase Storage

SYSTEM ARCHITECTURE

Client → **Flutter App** → **ASP.NET API** → **Supabase**

↓

Support Operator / Technician / Administrator

CLIENT USER JOURNEY

1. Open App
2. Select Device
3. Select Problem / Repair
4. Submit Request
5. Get Help / Track Status

IS IT VIABLE

Feasibility Study

✓ Technical
Feasible

Mature technologies, well-supported

✓ Operational
Feasible

Clear operational benefits

✓ Economic
Feasible

Free tools (R0 budget)

✓ Schedule
Feasible

Realistic timeline via MS Project

✓ Legal
Feasible

POPIA compliance with access controls

✓ Ethical
Feasible

Responsible use with clear policies

OVERALL DECISION: FEASIBLE WITH CHANGES

Must-Have core functions prioritised · Should-Have features (notifications, evidence) deferred · 24/7 monitoring treated as operational responsibility

WHAT COULD GO WRONG

Risk Analysis

TOP 3 PROJECT RISKS

#	Risk	Level	Score
1	**R01** **Clients may struggle to use the interface and submit support requests** *Mitigation: Keep the main client journey simple and easy to navigate*	High	9
2	**R08** **Limited development time may leave some requirements incomplete** *Mitigation: Prioritise Must-Have requirements before lower-priority features*	High	9
3	**R02** **Support requests may be missed or not handled on time** *Mitigation: Provide active monitoring and notifications where implemented*	Medium	6

10

Total Risks Identified

Every risk has an owner, mitigation, and contingency plan. Reviewed regularly with the Project Manager.

MS PROJECT

Project Schedule

PHASE 1 MILESTONES

Milestone	Date
Stakeholder Engagement Complete	25 Aug 2026
SRS, Feasibility, Risk & Schedule Drafted	28 Aug 2026
SRS Finalised (v2.0)	2 Sep 2026
Phase 1 Submitted	4 Sep 2026
Phase 1 Presented & Approved	8 Sep 2026

Today is 4 Sep 2026 — Phase 1 submission falls on schedule.

MAJOR PHASES

Phase 1: Engagement, Requirements & Feasibility

Stakeholder engagement, requirements, feasibility

24 Aug – 8 Sep 2026 · 12 days

Phase 2: Architecture, UML, ERD & Interface Design

System design, diagrams, prototype, stakeholder feedback

9 Sep – 23 Sep 2026 · 11 days

Phase 3: Development, Integration, Testing & Demo

Implementation, integration, testing, final demonstration

28 Sep – 13 Nov 2026 · 30 days

Schedule is realistic and on track

Tasks assigned to specific members · Dependencies defined · Progress tracked in the MS Project plan

REFINING THE PLAN

Key Decisions & Scope Changes

Simple client app → Non-Functional

Simplicity describes quality, not a business function

Notifications → Should Have

Lower priority than core request processing

24/7 management → Operational

Continuous staffing is business responsibility

Supporting evidence → Should Have

Useful but can be deferred

Asset / device info → Should Have

Relevant but lower priority

Laptop examples removed

Covered by problem / service selection

✓ **IN SCOPE**

- Client registration and login
- Device selection
- Problem / service selection
- Support-request submission
- Request status tracking
- Client / support communication
- Technician assignment
- Administrator controls
- Device / asset info (where required)
- Supporting evidence (where retained)

✕ **OUT OF SCOPE**

- Unapproved device categories
- Complex financial / accounting functions
- Advanced AI diagnosis
- Automated physical repair of devices
- Unauthorised access to client devices
- Features outside agreed requirements (without change control)

Team Contributions

All 10 members contributed across stakeholder engagement, requirements, feasibility, risk, scheduling, and documentation.

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PHASE 1 WRAP-UP

Conclusion & Q&A

✓ Phase 1 Completed Successfully

- Stakeholder identified and engaged (Prince Morafo)
- Requirements documented and confirmed (11 FRs, 5 NFRs)
- Feasibility study completed (6 areas)
- Risk register created (10 risks)
- Project schedule established (MS Project)
- GitHub repository active
- Progress tracker maintained
- All 10 members contributed

NEXT STEPS — PHASE 2

- System Design
- UML & ERD Diagrams
- Interface Design
- Prototype Development

Thank You

Questions & Answers

github.com/tmafunisa24-sudo/ITC327W-Project